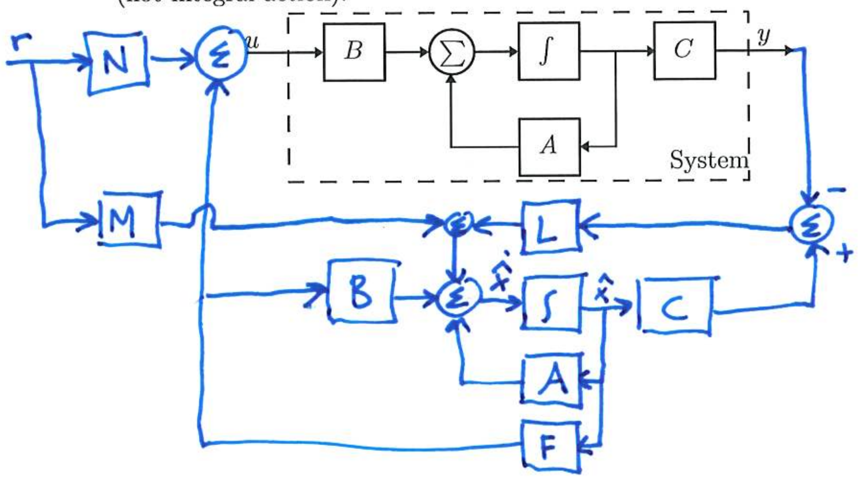
Test: Introducing Reference Signals

1. Complete the following diagram with a *full order observer*, a *state feedback*, and a *reference signal* (not integral action).



2. Given a system with poles at −1, −2, −3, and −4 (the open-loop system has no zeros). Where should two zeros be placed to maximize the bandwidth of the system?

Zeros should be placed at $s = -1$ and $s = -2$

3. Determine if the following system is controllable and observable, by calculating its controllability matrix and observability matrix.

$$\dot{x} = \begin{bmatrix} 1 & 0 \\ 1 & 2 \end{bmatrix} x + \begin{bmatrix} 1 \\ 0 \end{bmatrix} u$$
$$y = \begin{bmatrix} 1 & 0 \end{bmatrix} x.$$

$$\mathcal{O} = \begin{bmatrix} C \\ CA \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}$$

Not observable, since $\operatorname{rank} \mathcal{O} = 1$

$$\mathcal{C} = \begin{bmatrix} B & AB \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}.$$

Controllable, since $\operatorname{rank} \mathcal{C} = 2$

4. Compute the zeros of the following system

$$\dot{x} = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix} x + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u$$
$$y = \begin{bmatrix} 0 & 1 \end{bmatrix} x.$$

The zeros are values of z where

$$\begin{vmatrix} A - zI & B \\ C & D \end{vmatrix} = 0$$

$$\begin{vmatrix} -z & 1 & 0 \\ -2 & -3-z & 1 \\ 0 & 1 & 0 \end{vmatrix} = \begin{vmatrix} -z & 1 \\ 0 & 1 \end{vmatrix} \cdot (-1) = z$$

In conclusion, a zero is in $s = 0$